The Genie in the Jmol: Your Wish is My Command!

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ST · OLAF COLLEGE

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Acknowledgments

This presentation is dedicated to George Hardgrove, Professor of Chemistry at St. Olaf College 1959-

whose fascination with all things crystallographic sparked my interest in structure determination and the beauty of crystallographic symmetry

Acknowledgments

Huge thanks to the Jmol User Group for so many great idea and encouragement over the past 18 years, just a few listed here

Showing 250 results of 32619

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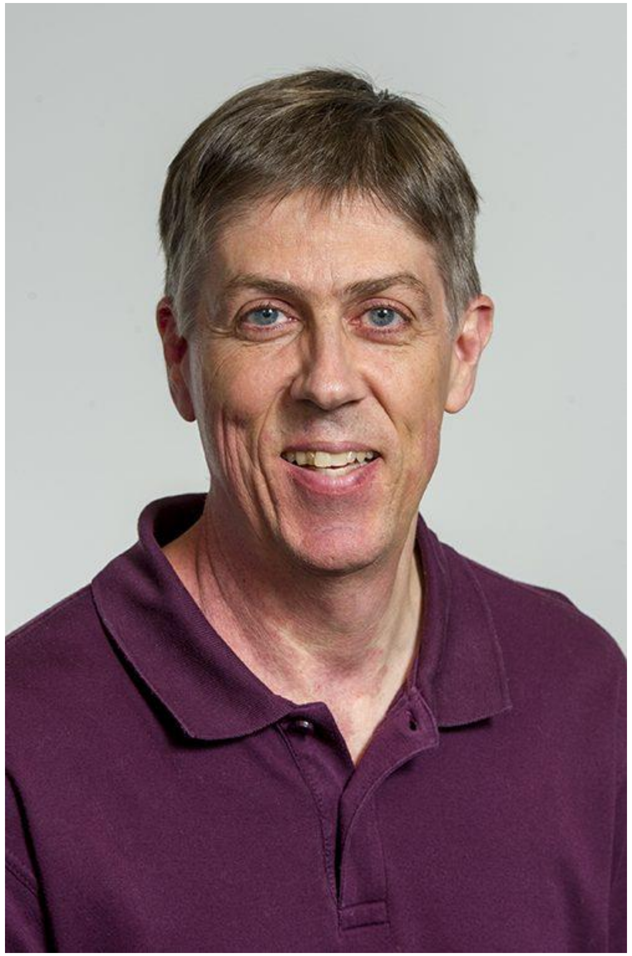
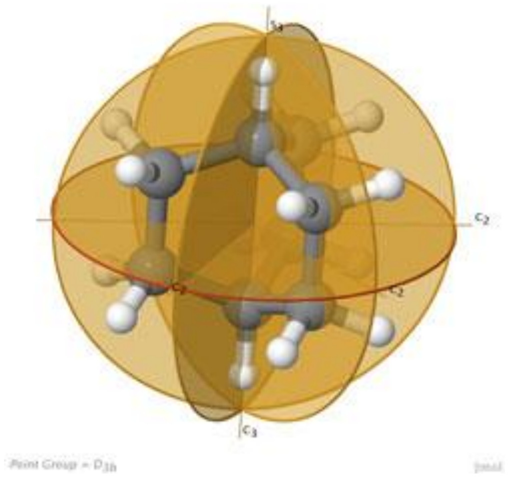
Geoge Whitwell
Alex Yang

Acknowledgments

Special thanks to **Dean Johnston** who, at a conference in July of 2008, gave me the idea of adding point group capability to Jmol and used it to develop the **world's best ever web site on point group symmetry**, https://symotter.org

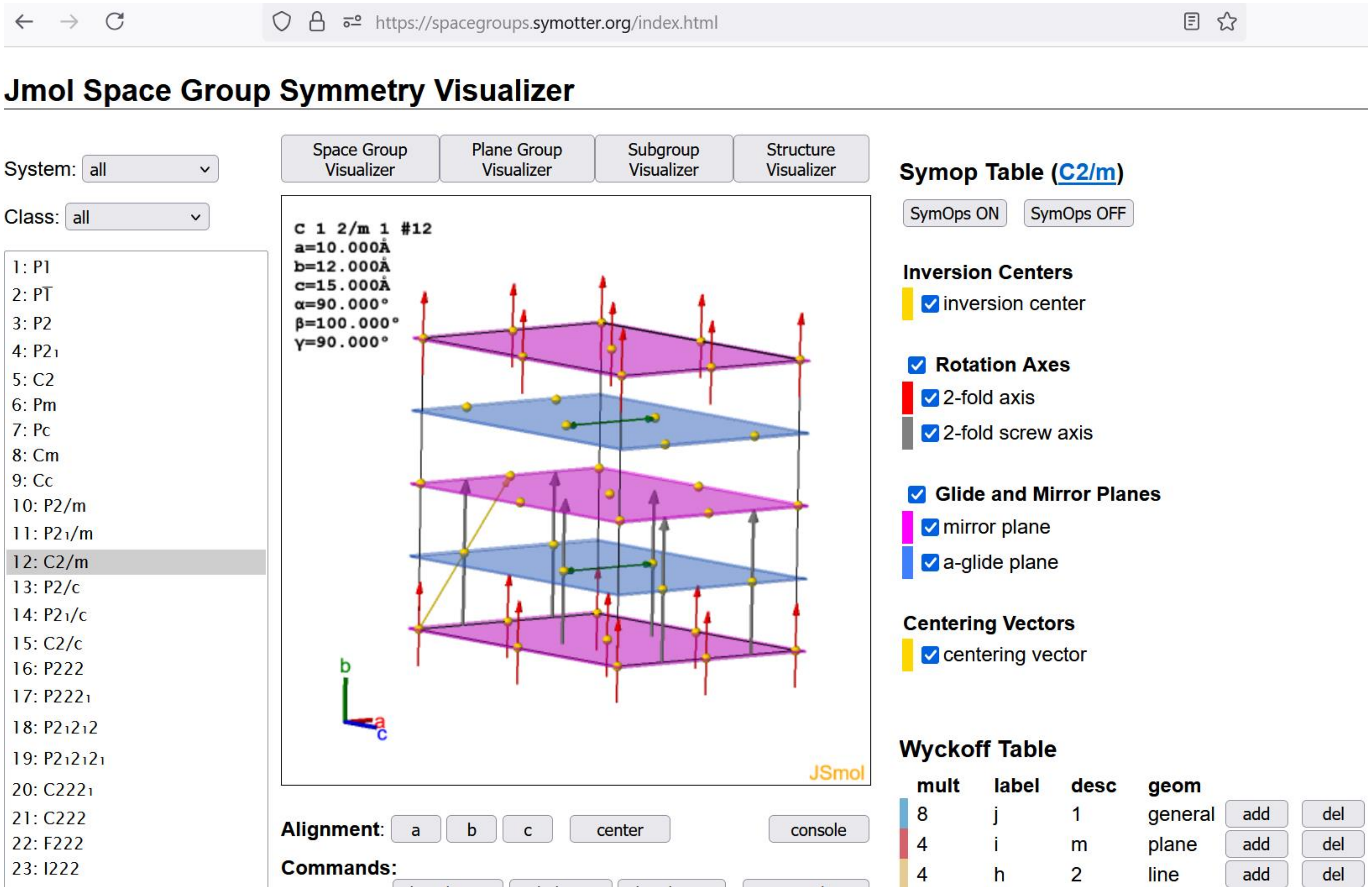




Recent Highlights

Created by Dean Johnston in 2024, this **killer webpage** implements many of the features of Jmol we will be discussing today.

https://spacegroups.symotter.org



Recent Highlights

Just published!

Bilbao Crystallographic Server, Cambridge Structural Database, *VESTA*, and *Jmol*.

teaching and education

JOURNAL OF APPLIED CRYSTALLOGRAPHY

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https://doi.org/10.1107/S1600576724007659

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Free tools for crystallographic symmetry handling and visualization

Gemma de la Flor,[a*] Mois I. Aroyo,[b] Ilaria Gimondi,[c] Suzanna C. Ward,[c] Koichi Momma,[d] Robert M. Hanson[e] and Leopoldo Suescun[f]

http://doi.org/10.1107/S1600576724007659

Recent Highlights

Just published!

Bilbao Crystallographic Server, Cambridge Structural Database, *VESTA*, and *Jmol*.

Table 1

Jmol `load` command options relating to crystallographic symmetry

Table 2

Jmol `modelkit` commands relating to crystallography

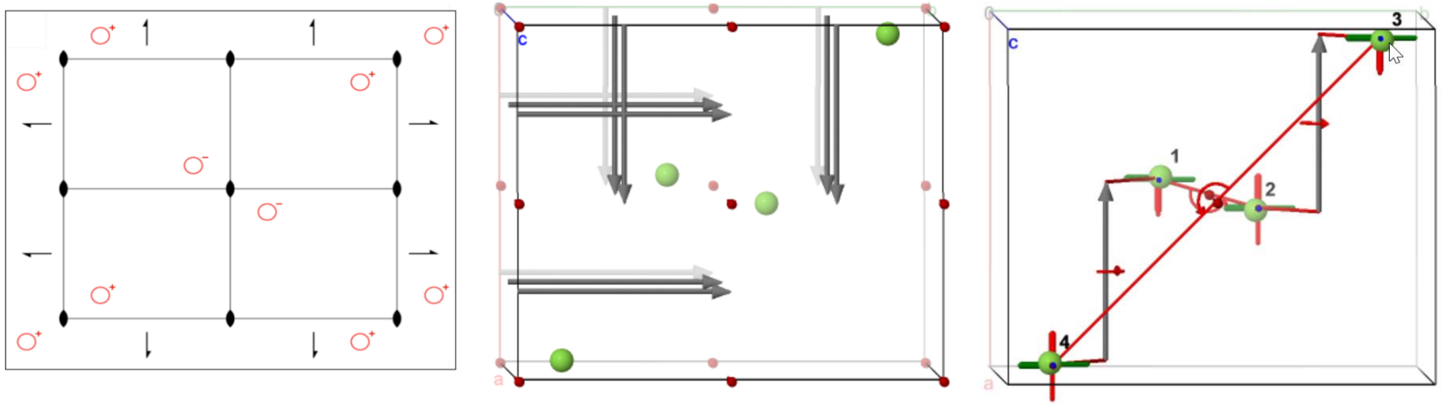
Table 3

Various questions and how to answer them using *Jmol* scripting functions

Table 4

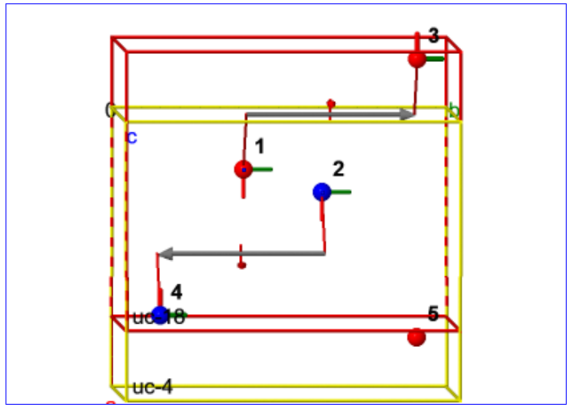
Representative websites incorporating *JSmol* in the area of crystallography

http://doi.org/10.1107/S1600576724007659



Recent Highlights

Includes a supporting website that allows you to experiment with Jmol and to **recreate all the figures in this section of the paper yourself** using JSmol, step by step.



NCI(small molecules) Search

open... console image display faster sharper large small
platformSpeed: 8 7 6 5 4 3 2 1
info show clear hide
spacefill wire ball&stick cartoons fancy not flat
color atomno color cpk color structure
isosurface vdw off mep translucent opaque
labels on off echo larger smaller

Figure script "figure16.e2.png"

zap;modelkit spacegroup 18; rotate x 2; rotate y 2

Illustrating the a different transformation pathway from P 21 21 21 (No.18) to P 21 (No. 4).

- modelkit add "F" {0.460 0.383 0.288/1}
- draw uc18 unitcell color red "uc-18"
- modelkit spacegroup "18>a,b,c;1/4,0,0>4" packed
- translate y -5
- draw uc4 unitcell color yellow "uc-4"
- label "%[atomno]";color labels black
- set labelOffset 4 4;font label 16 bold
- color property site
- draw s23 symop @1 @3
- draw s41 symop @2 @4
- set frank off;set showunitcellinfo false
- set picking dragatom
- prompt "BCS data used: " + spacegroup(18,4, "subgroups")

more examples

zap;modelkit spacegroup 18 Illustrating the differences in symmetry between P 21 21 21 (No. 18) and its subgroup P 21 (No. 4).

help Execute

https://chemapps.stolaf.edu/jmol/jsmol/iucrdemo

Quick Introduction to Jmol

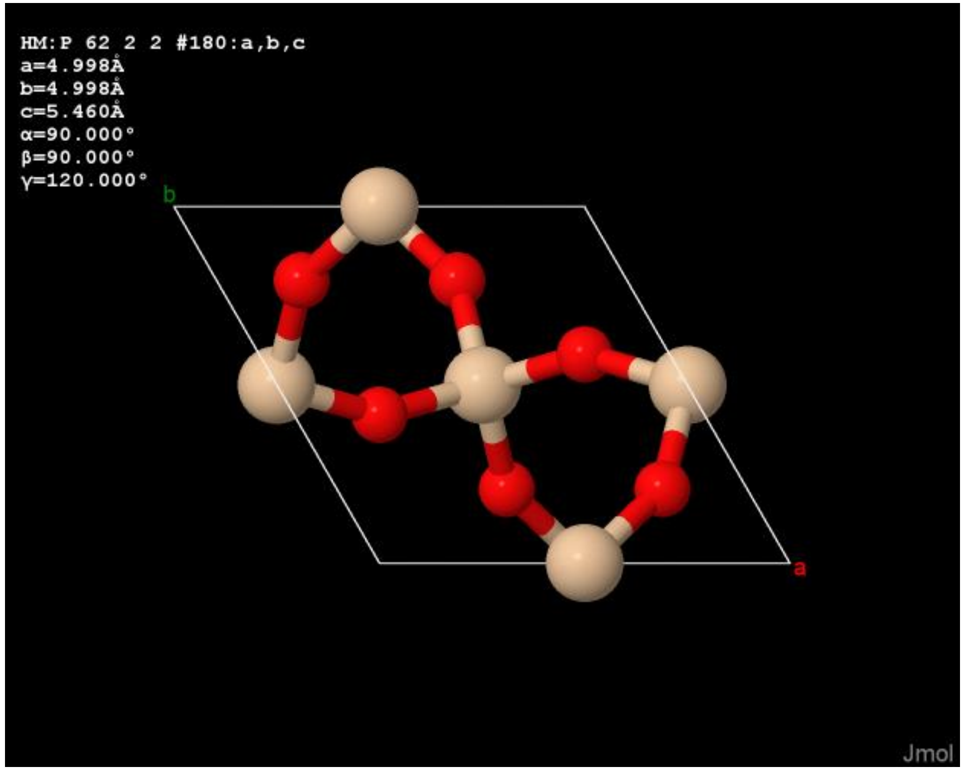
Mission: The high-quality, real-time visualization **and analysis** of molecular structure, dynamics, and energetics.

Jmol is:

- open source Java
- automatically transpiled to JavaScript
- cross-disciplinary
- actively being developed

Jmol is not:

- a commercial enterprise
- a quantum computational package



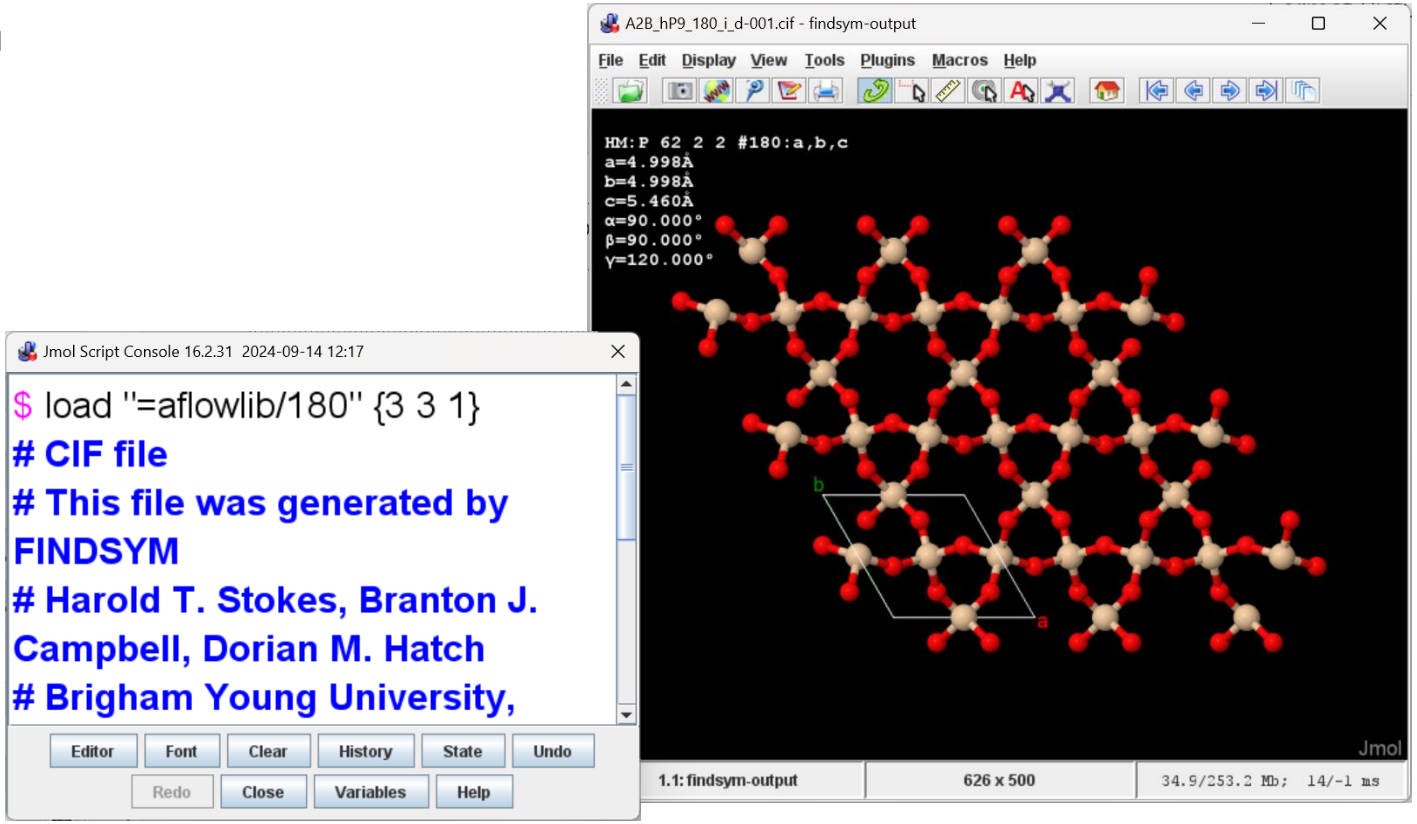
Quick Introduction to Jmol

Configurations:

Jmol.jar Java Application

fast, full service,

Allows drag&drop

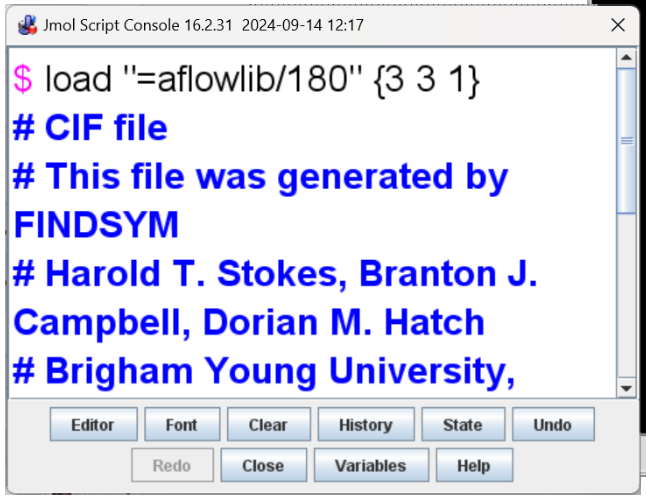


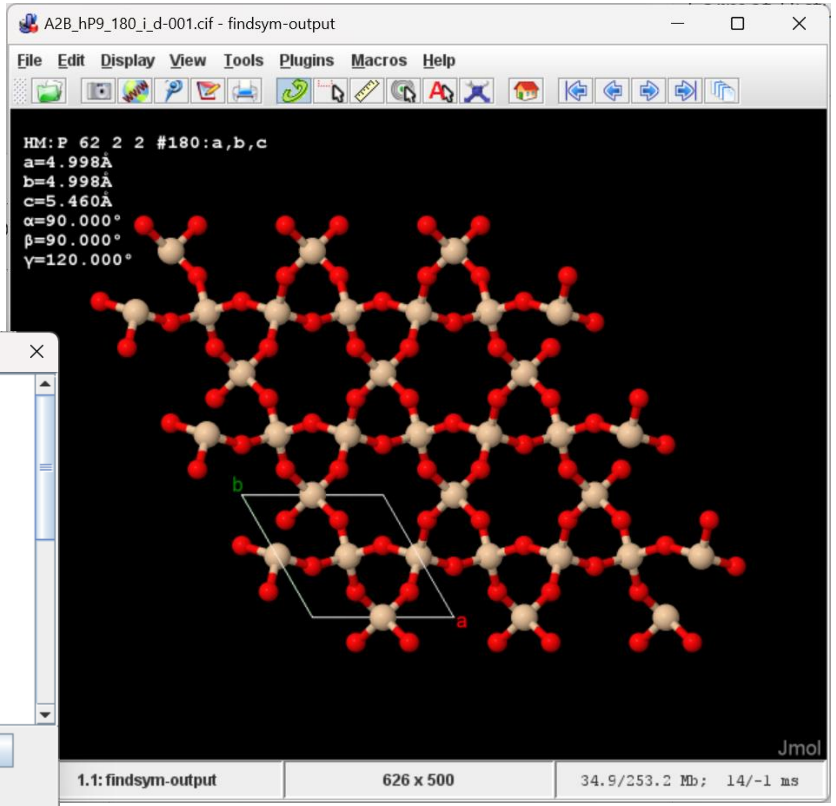
Quick Introduction to Jmol

Configurations:

JmolD.jar Java Application

Full double precision





Quick Introduction to Jmol

Configurations:

JmolData.jar and JmolDataD.jar

command-line interface

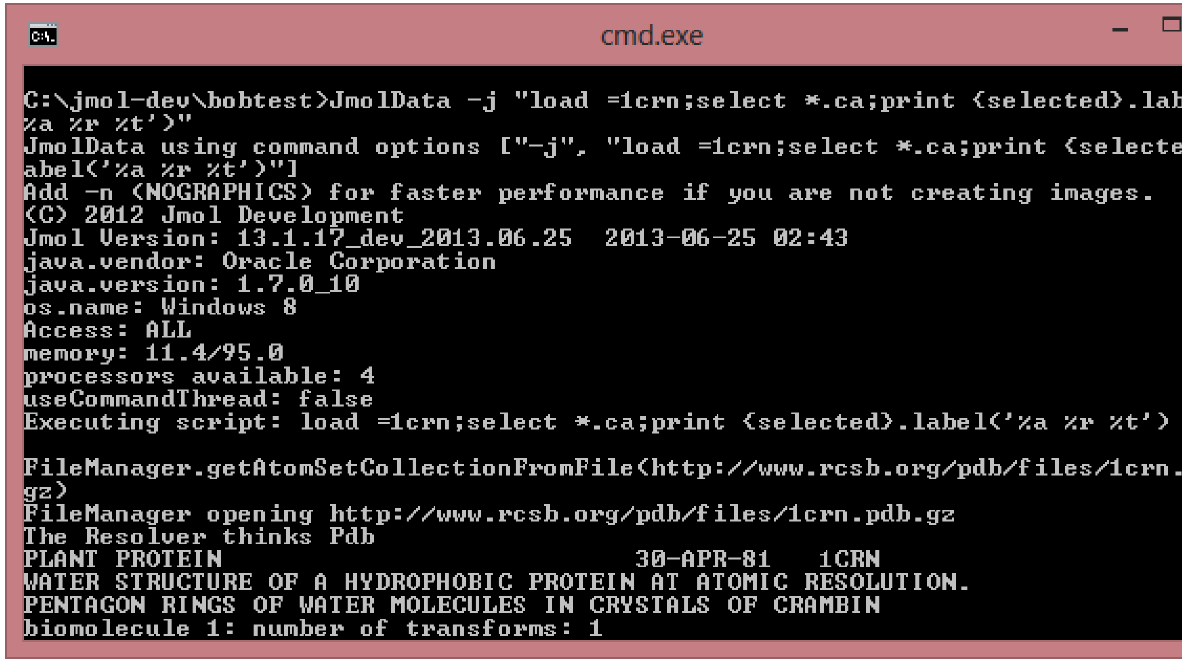
fully scriptable

runs “headless” (no user interface)

can create images and do analyses

allows for automated workflow

can be used as server-side app



Quick Introduction to Jmol

Configurations:

JavaScript App

double precision

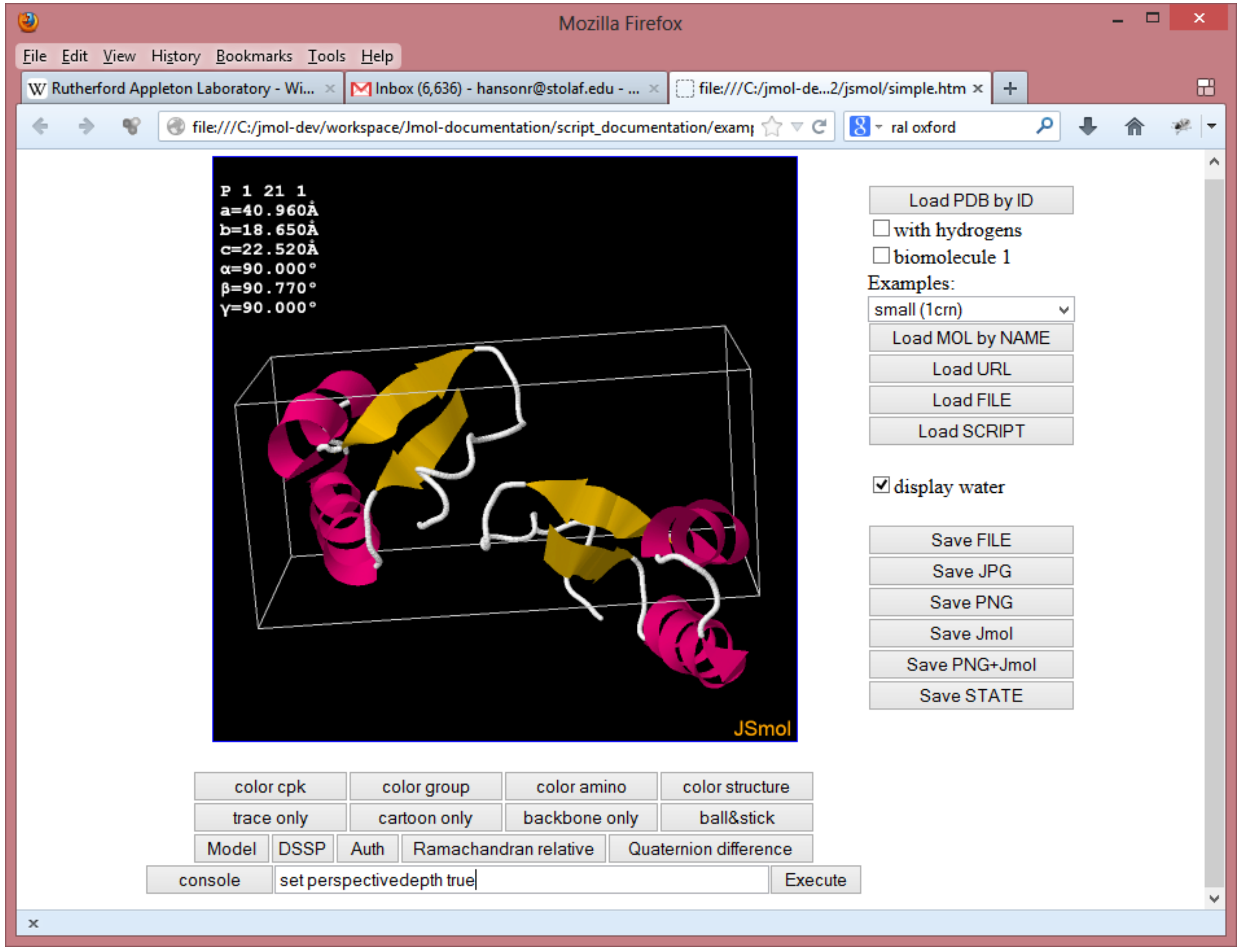
identical to JmolD.jar

works in all popular browsers, including

Safari for the iPad, android phones

allows drag&drop

somewhat slower than Java



I wish…

I wish…

Jmol - hansonr@stolaf.edu

Idea:

Name:

Email:

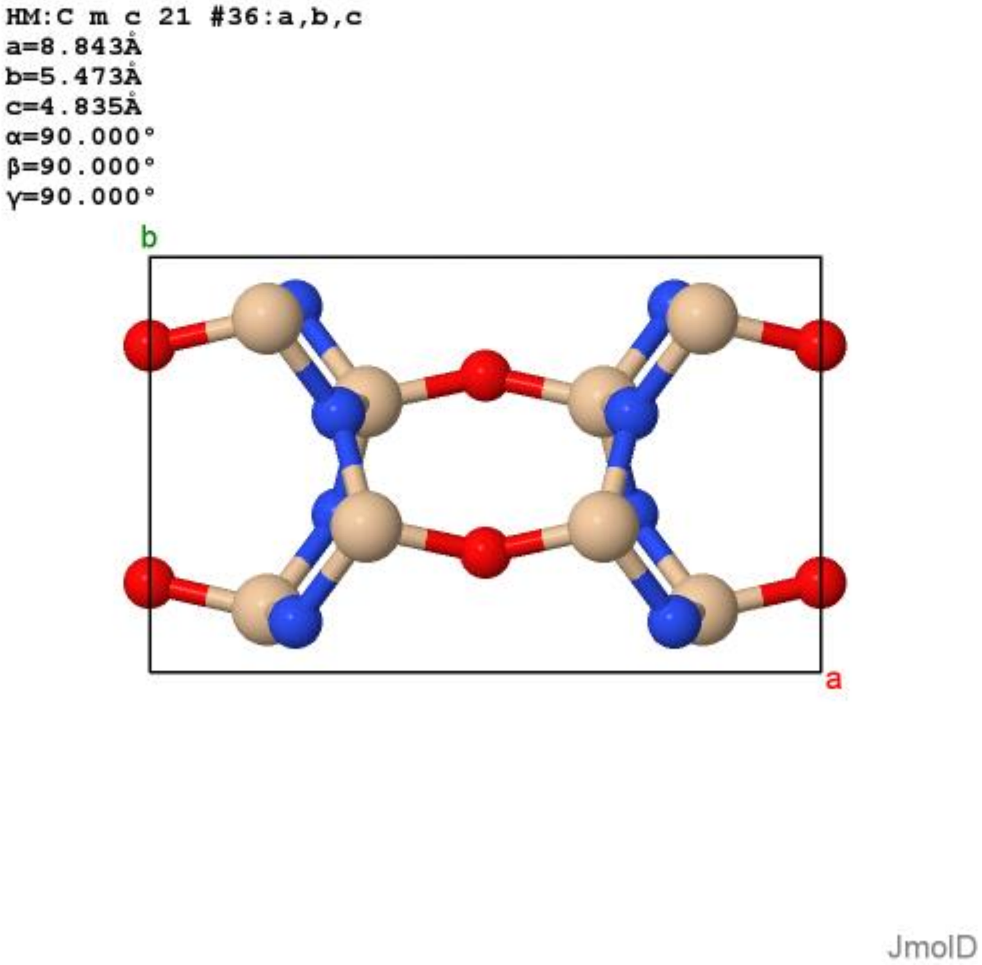
I wish…

…Jmol could quickly load example files from any space group.

I wish…

…Jmol could quickly load example files from any space group.

load =aflowlib/36.6 packed



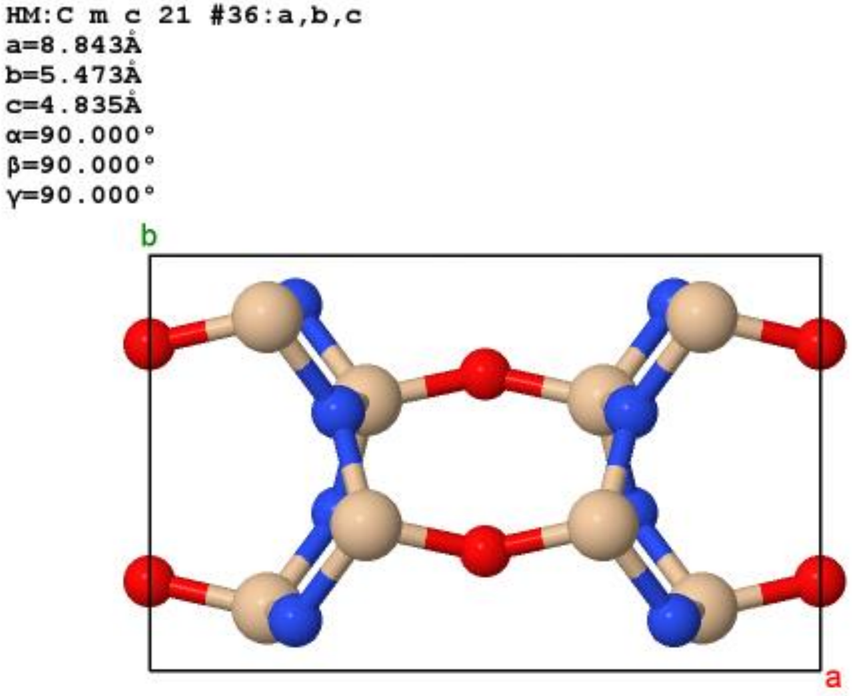
Loads a structure from the AFLOW Encyclopedia of Crystallographic Prototypes

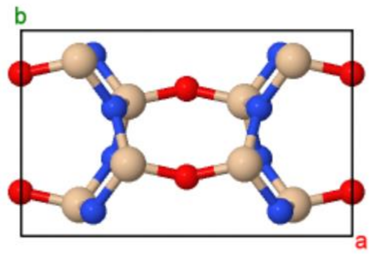
All space groups are represented by at least one structure.

https://aflowlib.org/prototype-encyclopedia

I wish…

…I could build a crystal from scratch using Jmol.





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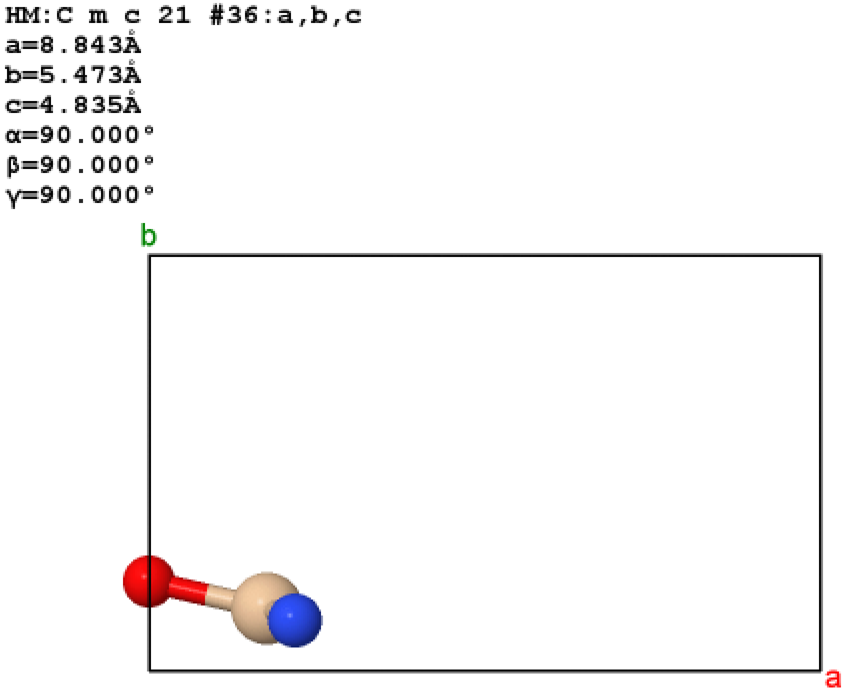


load =aflowlib/36.6

show unitcell

a=8.843, b=5.473, c=4.835,

alpha=90.0, beta=90.0, gamma=90.0

print {*}.label("%e %6.3[fxyz]")

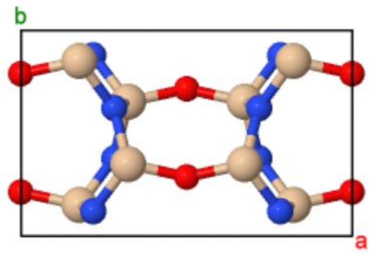
O 0.000 0.214 0.230

N 0.218 0.121 0.642

Si 0.176 0.151 0.290

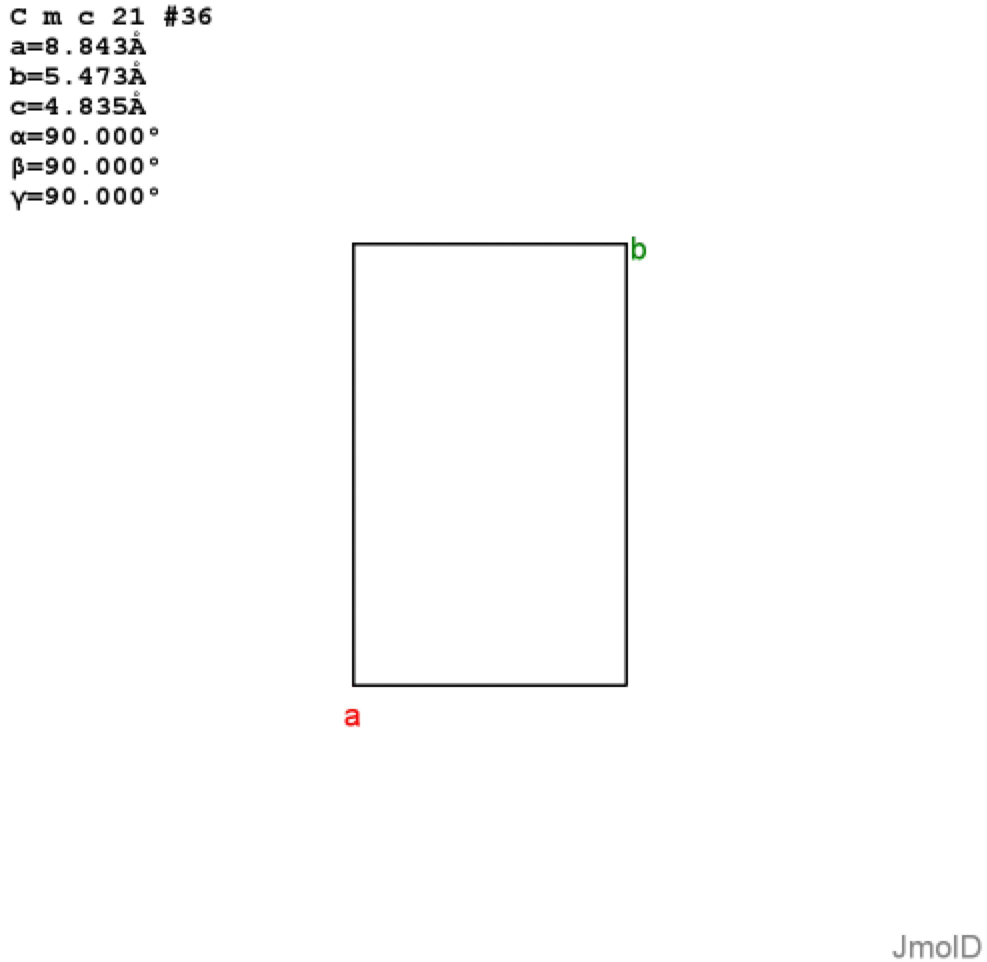
I wish…

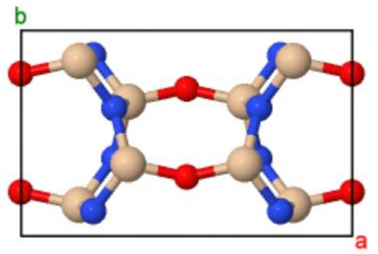
…I could build a crystal from scratch using Jmol.



modelkit zap spacegroup 36

unitcell [8.843 5.473 4.835 90 90 90]





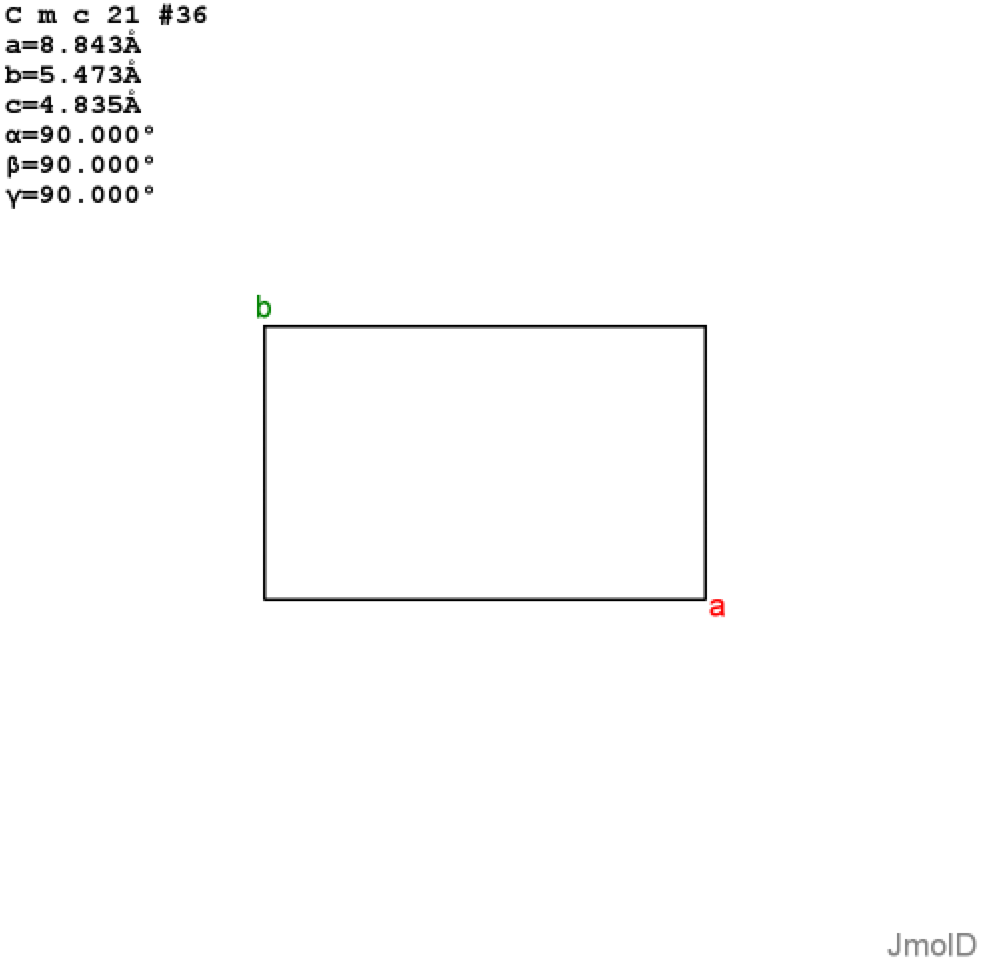
I wish…

…I could build a crystal from scratch using Jmol.

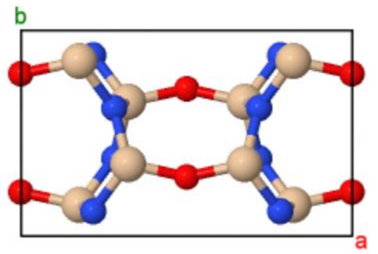
modelkit zap spacegroup 36

unitcell [8.843 5.473 4.835 90 90 90]

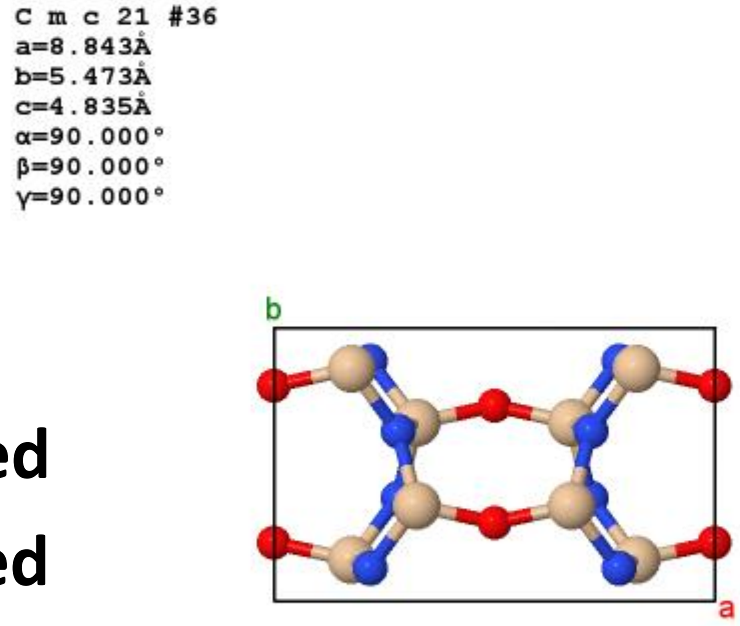
moveto axis c4



I wish…



…I could build a crystal from scratch using Jmol.



modelkit zap spacegroup 36

unitcell [8.843 5.473 4.835 90 90 90]

moveto axis c4

modelkit add O {0.000 0.214 0.230/1} packed

modelkit add N {0.218 0.121 0.642/1} packed

modelkit add Si {0.176 0.151 0.290/1} packed

connect

I wish…

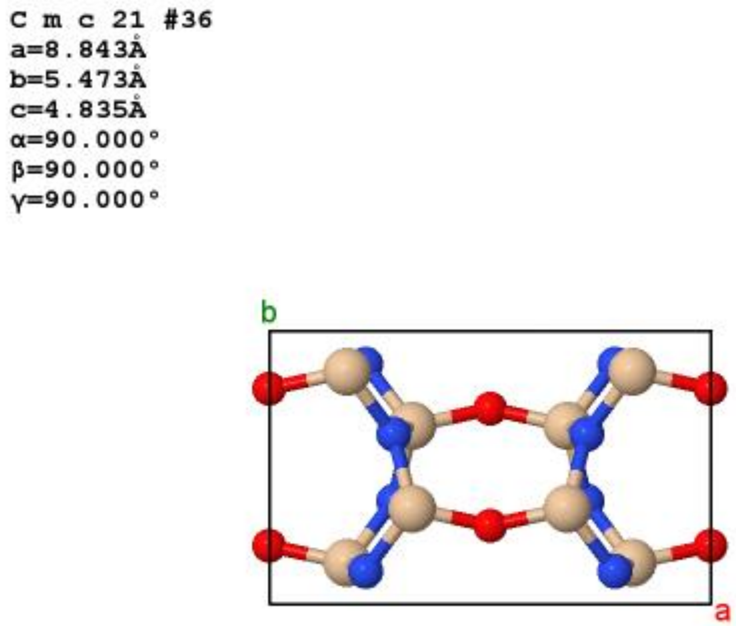
…I could move atoms around in a crystal structure.

I wish…

…I could move atoms around in a crystal structure.

load =aflowlib/36.6 packed

set picking dragatom

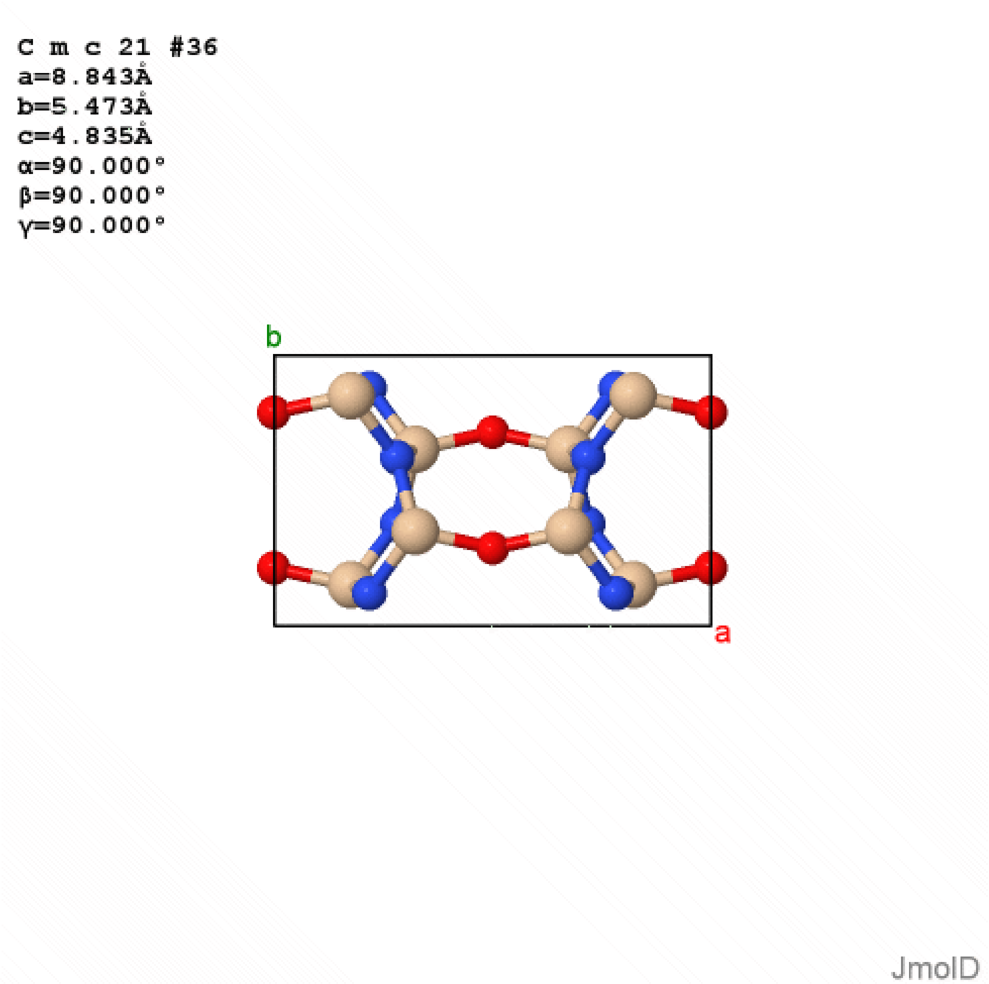


I wish…

…I could move atoms around in a crystal structure.

load =aflowlib/36.6 packed

set picking dragatom



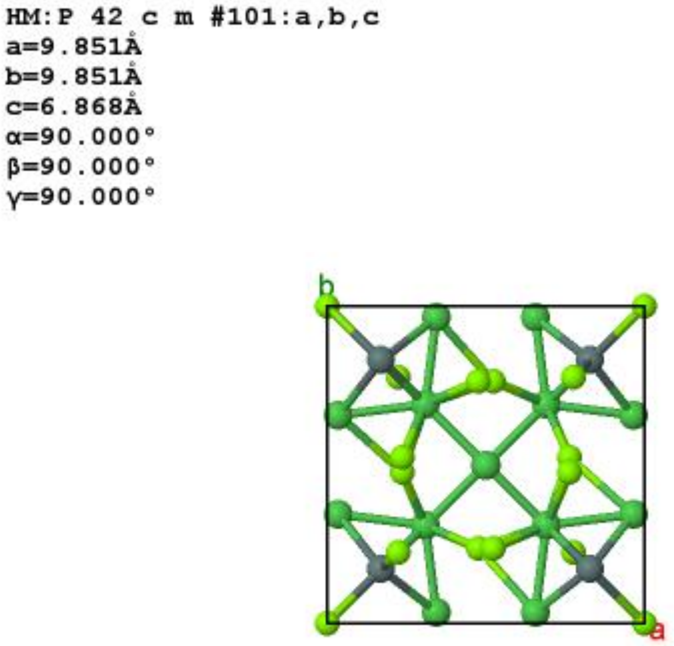
I wish…

…Jmol could switch among space group settings for a loaded model.

I wish…

…Jmol could switch among space group settings for a loaded model.

load =aflowlib/101 packed;zoom 60



JmolD

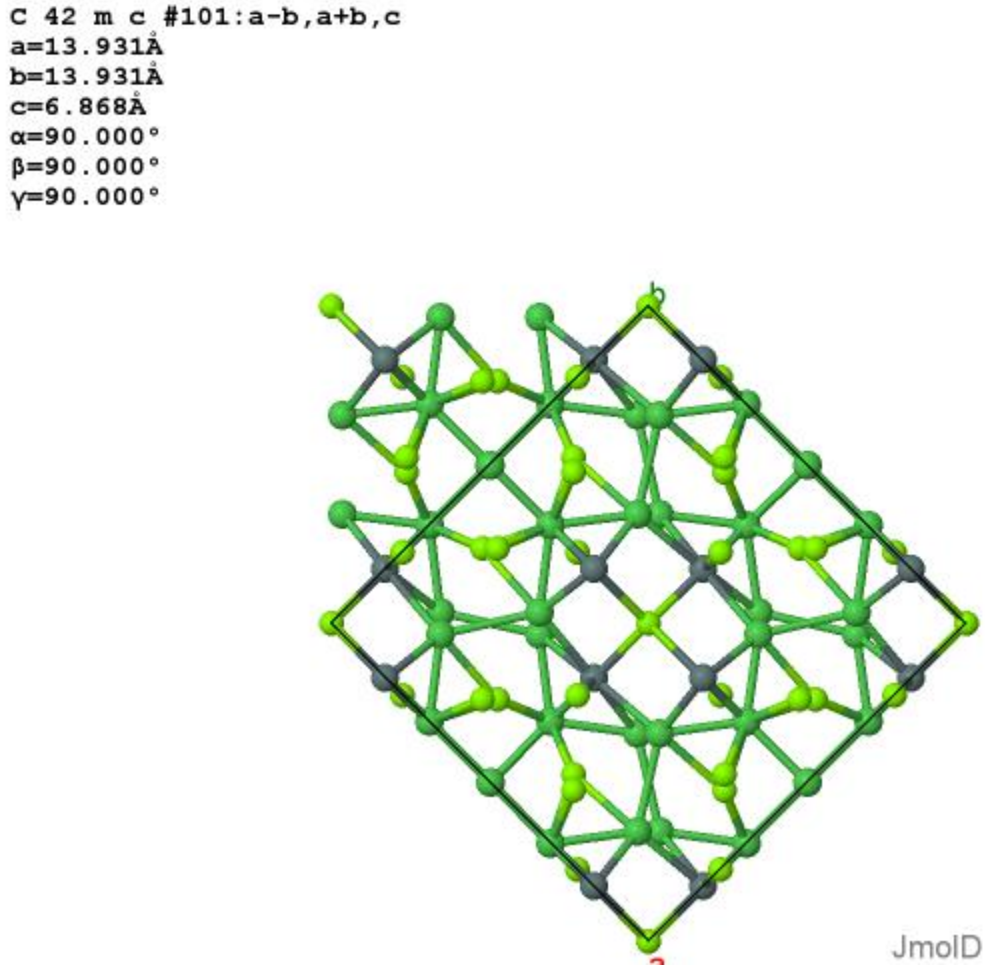
I wish…

…Jmol could switch among space group settings for a loaded model.

load =aflowlib/101 packed;zoom 60

modelkit spacegroup "C 42 m c" packed

connect



I wish…

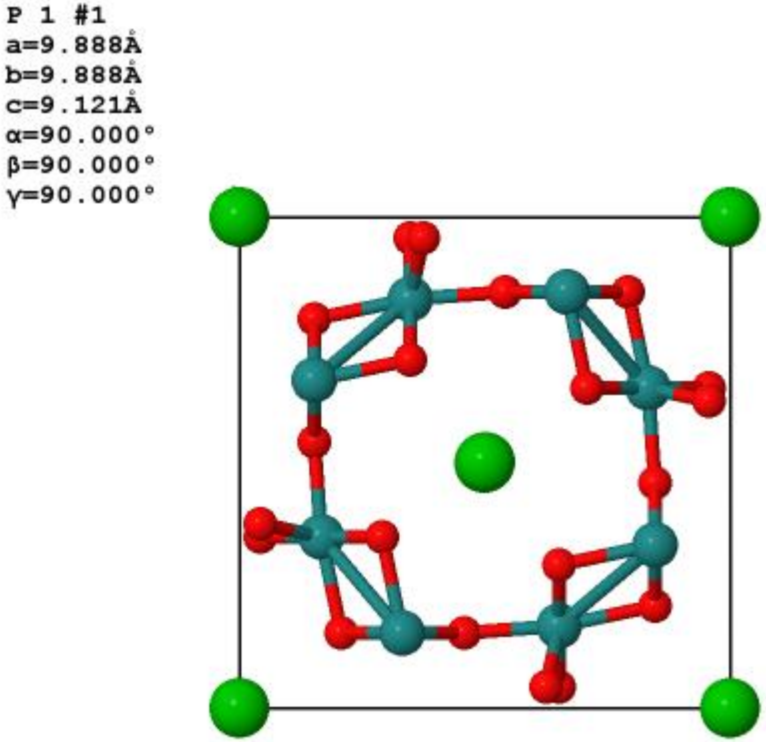
…Jmol could identify a model’s space group based on its symmetry.

I wish…

…Jmol could identify a model's space group based on its symmetry.

load =aflowlib/75 packed

modelkit spacegroup p1



I wish…

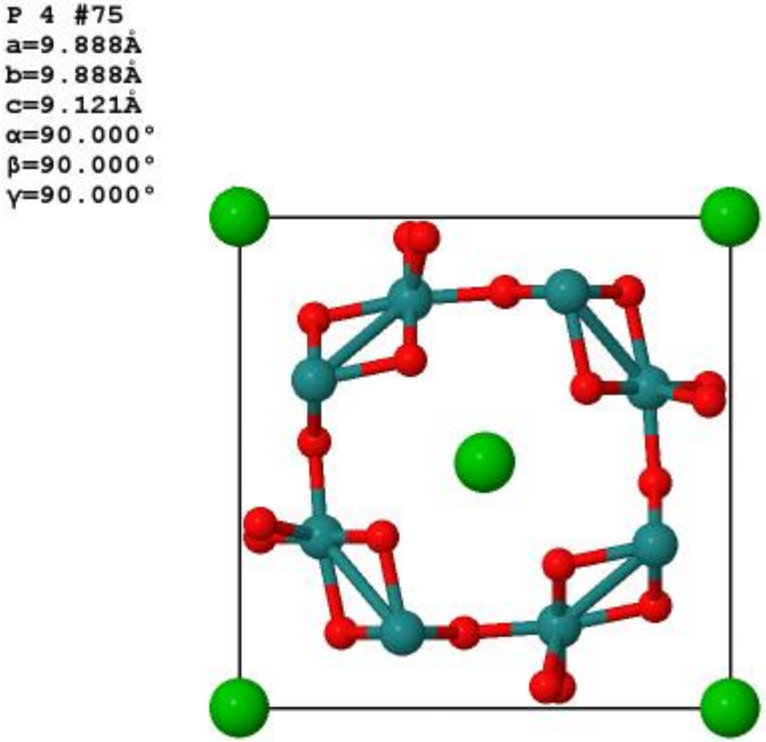
…Jmol could identify a model’s space group based on its symmetry.

load =aflowlib/75 packed

modelkit spcaegroup p1

modelkit spacegroup

75 HM:P 4 #75:a,b,c



I wish…

…I could show a space group's symmetry elements in Jmol.

I wish…

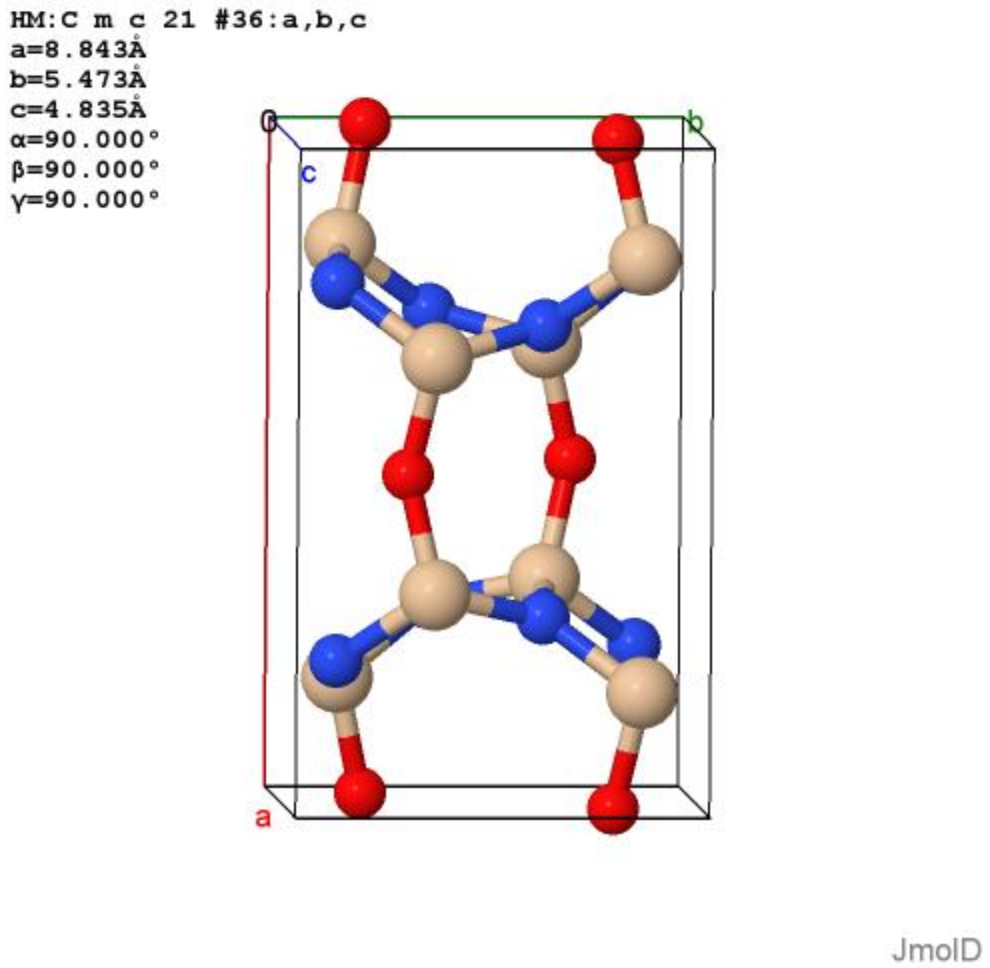
…I could show a space group's symmetry elements in Jmol.

load =aflowlib/36.6 packed

moveto axis c1

rotate x 5

rotate y 5



I wish…

…I could show a space group's symmetry elements in Jmol.

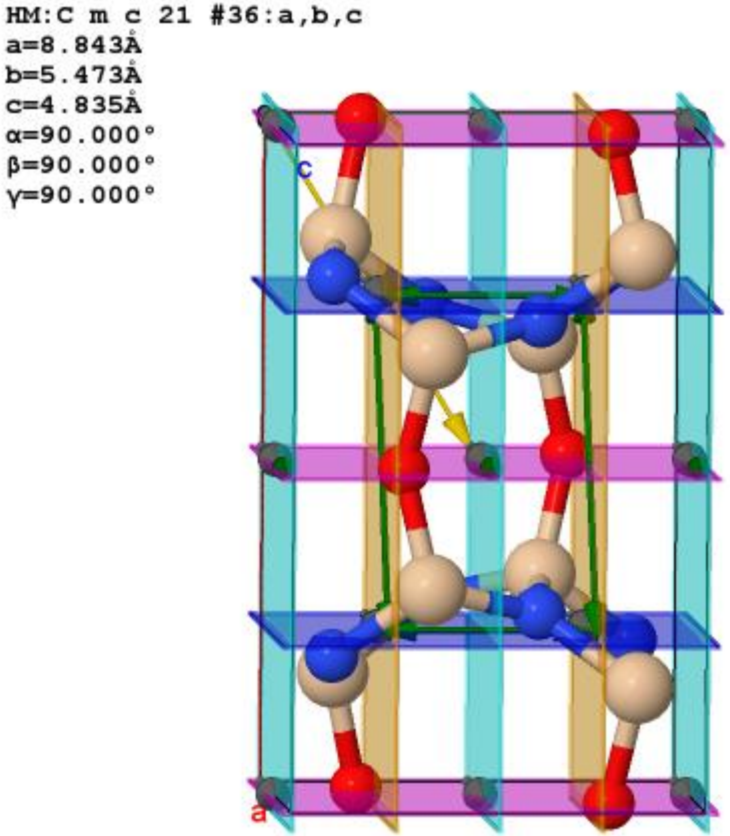
load =aflowlib/36.6 packed

moveto axis c1

rotate x 5

rotate y 5

draw spacegroup all



I wish…

…I could show just ***some*** of a space group's symmetry elements in Jmol.

I wish…

…I could show just ***some*** of a space group’s symmetry elements in Jmol.

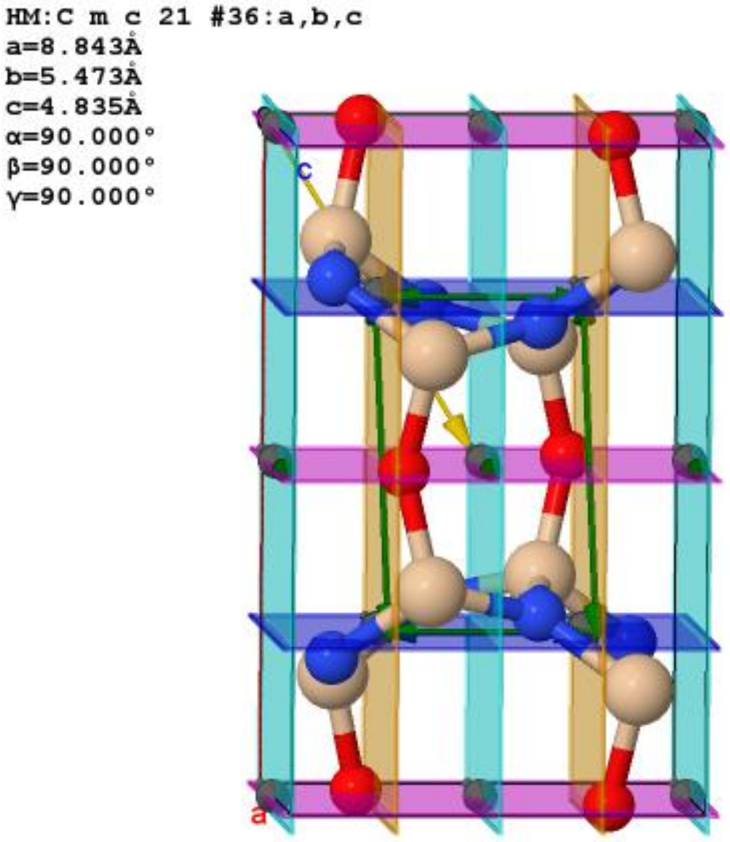
load =aflowlib/36.6 packed

moveto axis c1

rotate x 5

rotate y 5

draw spacegroup all

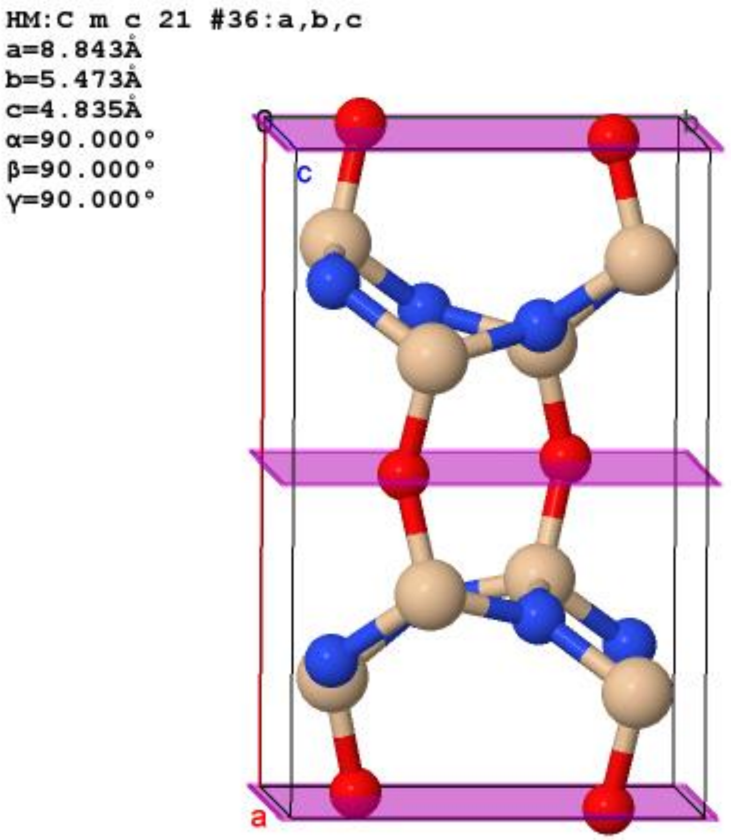


I wish…

…I could show just ***some*** of a space group's symmetry elements in Jmol.

draw off

draw *mirror* on



JmolD

I wish…

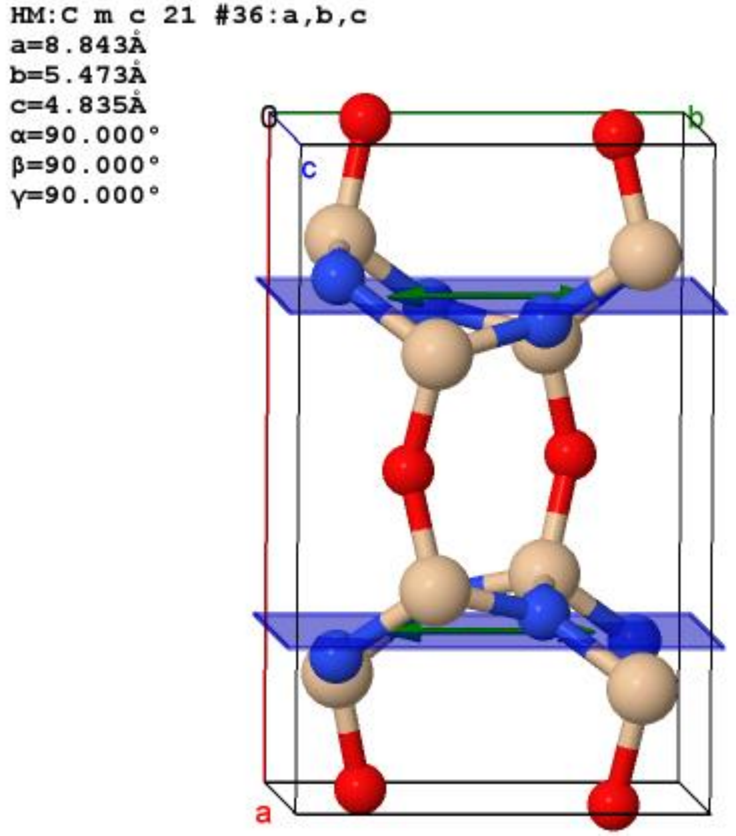
…I could show just ***some*** of a space group's symmetry elements in Jmol.

draw off

draw *mirror* on

draw off

draw *b-g* on



JmolD

I wish…

…I could show just ***some*** of a space group's symmetry elements in Jmol.

draw off

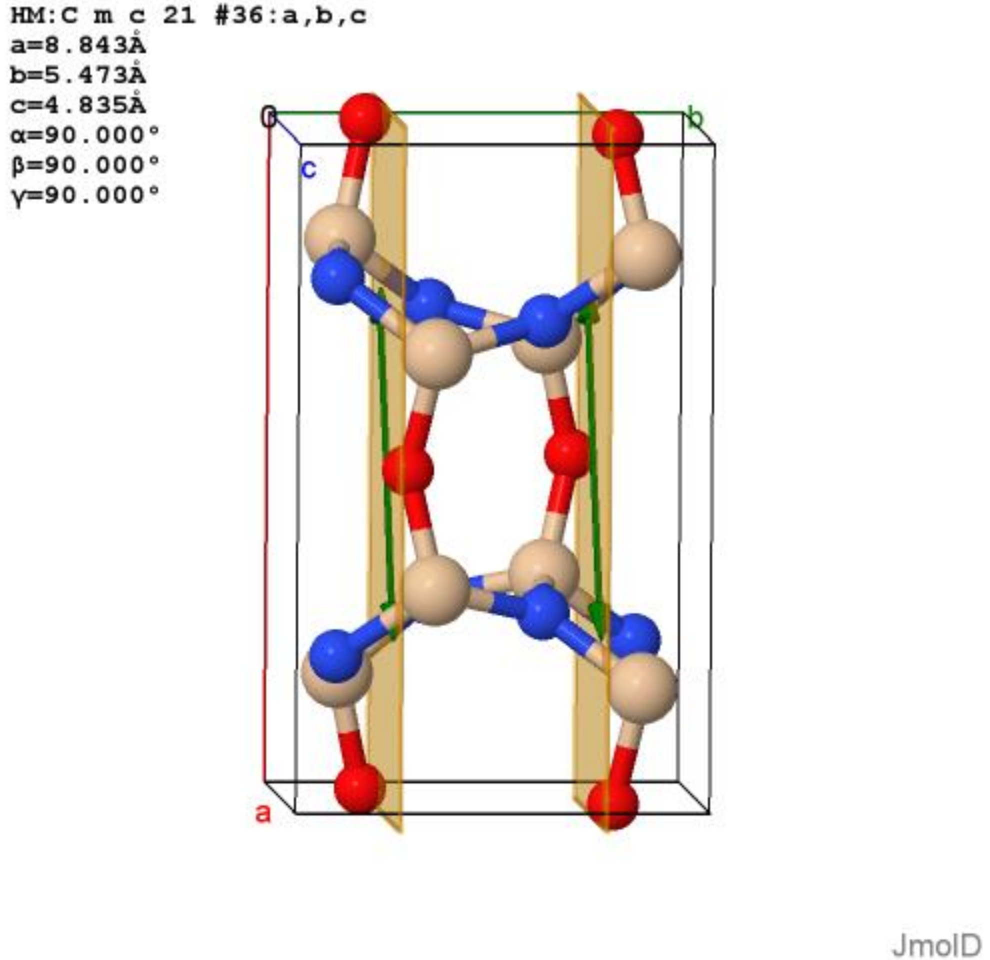
draw *mirror* on

draw off

draw *b-g* on

draw off

draw *n-g* on



I wish…

…I could show just ***some*** of a space group's symmetry elements in Jmol.

draw off

draw *mirror* on

draw off

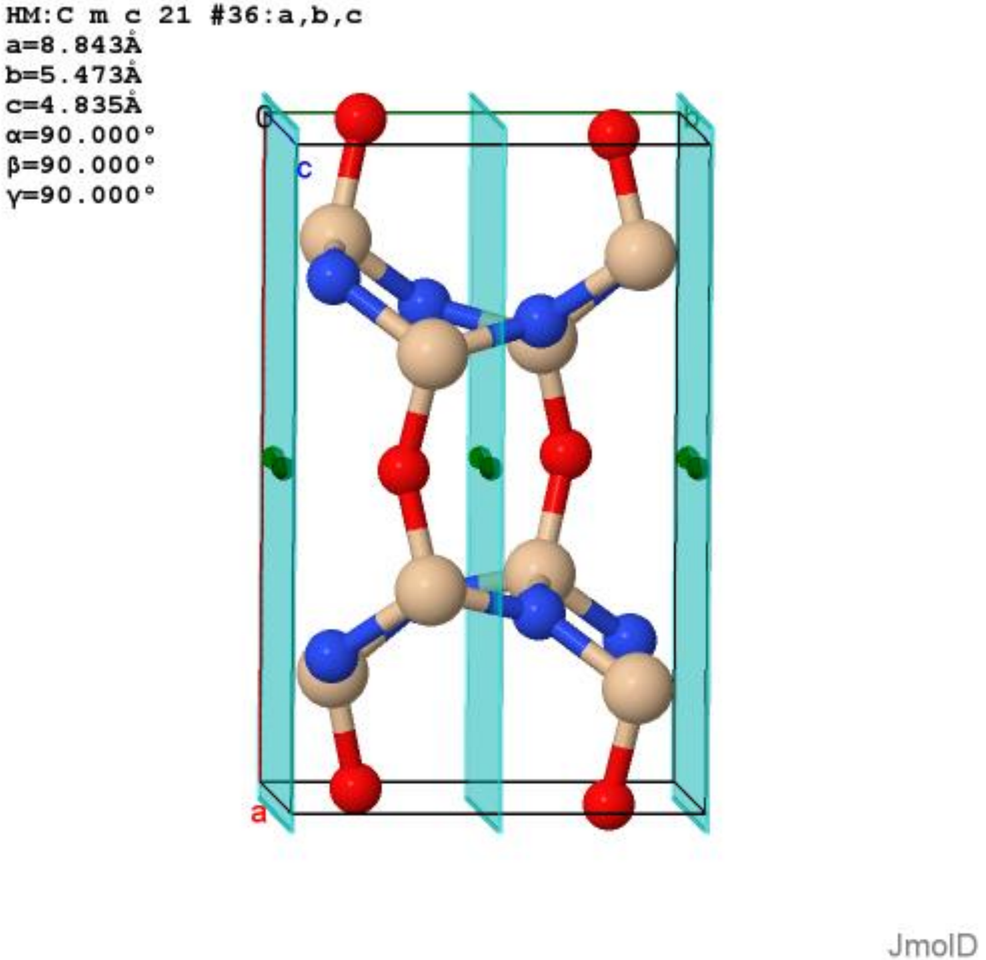
draw *b-g* on

draw off

draw *n-g* on

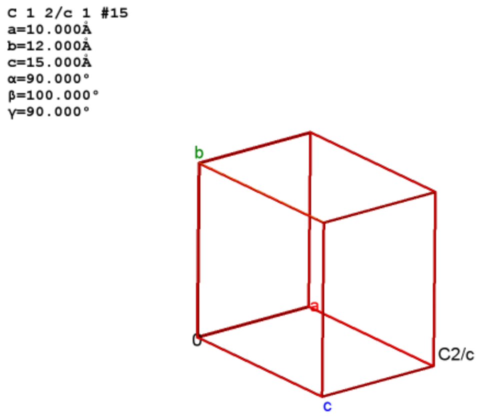
draw off

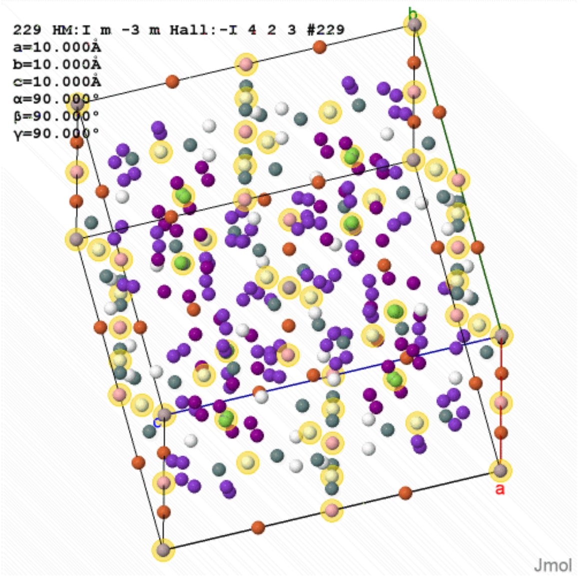
draw *c-g* on



I wish…

…Jmol could…





?

JmolD

Your turn!

∞

https://chemapps.stolaf.edu/presentations/laca2024